



Innovations in fish freshness using Deep Learning

Comparative analysis with existing patents and research

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Date: 17-06-2024



Introduction

- Importance of automated fish freshness assessment in the food industry
- Objective: Comparative analysis of existing patents and research in fish freshness classification using deep learning



Paper 1 - Framework for Fish Freshness Detection (Sciencedirect, 2024)

Method: Mask R-CNN with robotic arm and fisheye analysis

Dataset: 5000 fish eye images, species include Shrimps, Ilish, Tilapia, Pangas, Rohu, Magur

Accuracy: 96.5%

Novelty: Integration of robotic arm for fish removal, Mask R-CNN for pixel-level segmentation

Link: <https://www.sciencedirect.com/science/article/pii/S2666154324001765>



Paper 2 - Prediction Method of Large Yellow Croaker Freshness (Springer, 2024)

Method: Improved Residual Neural Network (ResNet-34)

Dataset: 100 images, based on TVB-N value over 9 days

Accuracy: 99.4%

Novelty: Focus on pupil feature extraction using mixed attention mechanism

Link: <https://link.springer.com/article/10.1007/s11694-024-02381-5>



Paper 3 - Detection of Fish Freshness using AI Methods (Springer, 2023)

- **Methods:** SqueezeNet, InceptionV3
- **Dataset:** 4476 fish body images
- **Accuracy:** 100% using SVM, ANN, LR
- **Novelty:** Objective freshness assessment without gill inspection
- **Link:** <https://link.springer.com/article/10.1007/s00217-023-04271-4>



Paper 4 - Image Processing for Fish Quality Assessment (Sciencedirect, 2016)

Method: Wavelet domain analysis for gill tissue segmentation

Species: Rohu

Novelty: Automatic segmentation using wavelet coefficients, non-destructive quality assessment.

Link: <https://www.sciencedirect.com/science/article/abs/pii/S0260877415300881>



Paper 5 - Fisheye Freshness Detection with Mobile App (Springer, 2024)

Method: VGG19 with basic ML models (KNN, LR, DT, RF, SVM, ANN)

Dataset: FFE dataset

Novelty: Mobile application integration for real-time freshness assessment

Link: <https://link.springer.com/article/10.1007/s00217-024-04493-0>



Comparative Analysis

Paper	Method	Dataset Size	Species	Accuracy	Novelty
Paper 1	Mask R-CNN	5000	Shrimps, Ilish, Tilapia, etc	96.5%	Robotic arm integration
Paper 2	ResNet-34	100	Large Yellow Croaker	99.4%	Pupil-focused feature extraction using mixed attention mechanism
Paper 3	SqueezeNet, InceptionV3	4476	Not specified	100%	Objective freshness assessment without gill inspection
Paper 4	Wavelet domain analysis	Not specified	Rohu	Not specified	Automatic segmentation using wavelet coefficients
Paper 5	VGG19 with ML models	Not specified	Not specified	77.3%	Mobile application integration



Innovation Highlights

Key innovations across the reviewed papers:

- Robotic arm integration (Paper 1)
- Pupil-focused freshness assessment (Paper 2)
- Non-invasive freshness evaluation (Paper 3)
- Wavelet-based gill tissue segmentation (Paper 4)
- Mobile application for on-the-go assessment (Paper 5)

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- **Paper 1:** Utilizes Mask R-CNN for fish freshness detection with a robotic arm setup.
 - **Paper 2:** Applies ResNet-34 with attention mechanisms for freshness classification based on fish eye images.
 - **Paper 3:** Achieves 100% accuracy using deep learning (SqueezeNet, InceptionV3) for fish freshness assessment.
 - **Paper 4:** Focuses on wavelet domain analysis of fish gills for quality assessment.
 - **Paper 5:** Implements VGG19 for freshness detection via a mobile application.



Key Highlights of our fish freshness classification model

1. **Multi-Model Integration:** We utilize a YOLOv8-based fish detection model to identify and segment multiple fish within an image. Each detected fish undergoes further analysis through specialized models for cuts and damages detection and size-based classification.
2. **Logistic Regression Model:** For size-based classification, we employ a logistic regression model that uses fish dimensions to categorize them into good and bad quality, based on predefined thresholds.



3. Transfer Learning for Softness Assessment: A transfer learning model based on DenseNet121 evaluates fish softness, providing additional insights into freshness and quality.

4. Innovative Approach: Our system stands out due to its holistic approach, combining precise detection, damage assessment, and quality evaluation without the need for invasive methods like gill inspection, as seen in some existing solutions.



Conclusion

Summary of Findings:

- Diverse approaches to fish freshness detection using deep learning and image processing.
- High accuracy rates across different methods, highlighting the potential for reliable automated systems.
- Novel contributions include robotic arm integration, pupil-focused feature extraction, non-invasive freshness evaluation, wavelet-based gill tissue segmentation, and mobile app integration.



Implications

- The varied methods demonstrate the flexibility of deep learning in addressing the fish freshness classification challenge.
- Innovations such as robotic arm integration and mobile applications show the potential for practical, real-world implementations.



Next Steps

Patent Application:

- Explore patent application opportunities for the unique aspects of the proposed fish freshness classification system.
- Highlight the combination of YOLOv8 for detection, cuts and damages detection, and sequential classification models (logistic regression, transfer learning with DenseNet121).



Further Research

- Investigate the scalability of the proposed system for different fish species and larger datasets.
- Explore enhancements to the detection and classification models for improved accuracy and robustness.
- Consider real-world deployment scenarios and integration with existing fish processing workflows.